

BUFFALO NARROWS, SK, Canada

WMO: 710770

Lat: 55.8409N

Lon: 108.4208W

Elev: 440

StdP: 96.15

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.3	-32.4	-39.1	0.1	-35.0	-36.3	0.1	-32.2	10.4	-13.3	9.2	-12.8	1.8	300	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	27.3	17.6	25.4	16.7	23.8	16.0	18.9	24.8	18.0	23.4	17.2	22.1	3.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.6	21.0	16.0	12.0	20.0	15.2	11.4	19.2	55.6	24.9	52.5	23.4	50.0	22.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.1	7.1	-39.8	30.8	3.2	1.5	-42.1	31.9	-43.9	32.8	-45.7	33.6	-48.0	34.7		
			-39.8	20.9	3.2	1.0	-42.1	21.6	-44.0	22.2	-45.8	22.8	-48.2	23.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-17.0	-14.4	-7.3	1.8	9.3	15.2	17.9	16.6	11.0	3.2	-6.9	-14.6	(6)
(7)		DBStd	13.88	9.39	8.11	8.54	5.94	4.91	3.45	2.67	3.30	4.22	4.72	6.66	8.62	(7)
(8)		HDD10.0	3906	837	682	538	249	73	4	0	1	38	215	507	762	(8)
(9)		HDD18.3	6267	1096	916	796	495	281	104	39	73	221	469	757	1021	(9)
(10)		CDD10.0	739	0	0	0	4	52	159	246	205	68	5	0	0	(10)
(11)		CDD18.3	57	0	0	0	0	1	9	27	19	1	0	0	0	(11)
(12)		CDH23.3	461	0	0	0	1	23	91	193	136	18	0	0	0	(12)
(13)		CDH26.7	82	0	0	0	0	3	17	33	26	2	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.0	3.0	3.2	3.3	3.1	3.0	3.1	3.4	3.6	3.4	3.0	(14)	
(15)	Precipitation	PrecAvg	460	25	18	18	28	36	69	80	61	47	30	23	23	(15)
(16)		PrecMax	636	103	49	36	66	73	202	155	175	123	80	65	51	(16)
(17)		PrecMin	262	7	1	1	4	1	20	28	2	14	0	7	6	(17)
(18)		PrecStd	89	18	12	9	16	19	37	32	35	21	19	14	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	6.6	11.2	20.9	26.3	28.9	29.6	29.5	25.7	19.1	8.1	4.3	(19)
(20)			MCWB	2.5	2.6	4.9	9.9	14.2	17.1	19.1	18.4	16.7	12.0	4.3	1.6	(20)
(21)		2%	DB	1.7	2.2	8.2	16.4	22.8	25.6	27.4	26.5	22.4	14.6	5.2	1.1	(21)
(22)			MCWB	0.1	-0.5	3.2	8.1	12.7	15.6	18.2	17.8	15.1	9.2	2.6	-0.4	(22)
(23)		5%	DB	-1.0	-0.6	5.7	13.4	20.5	23.8	25.5	24.6	19.8	12.0	3.0	-0.9	(23)
(24)			MCWB	-2.1	-2.1	1.9	6.5	11.5	14.8	17.2	17.0	13.7	7.9	1.0	-1.9	(24)
(25)		10%	DB	-3.7	-2.8	3.5	10.8	18.0	21.8	23.8	22.8	17.7	9.7	1.2	-3.3	(25)
(26)			MCWB	-4.5	-3.9	0.6	5.1	10.5	14.1	16.7	16.3	12.7	6.4	0.0	-4.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.7	5.2	10.7	15.6	18.7	20.5	20.0	17.5	12.5	4.7	2.0	(27)
(28)			MCDWB	4.9	6.2	10.3	19.8	22.6	25.7	26.7	26.8	23.3	18.6	7.5	3.8	(28)
(29)		2%	WB	0.1	-0.2	3.4	8.3	13.9	17.2	19.2	18.7	15.8	10.0	2.8	-0.3	(29)
(30)			MCDWB	1.6	1.9	7.6	15.6	20.9	22.7	24.9	24.5	20.8	13.6	5.0	0.8	(30)
(31)		5%	WB	-2.1	-2.0	2.1	6.8	12.5	16.2	18.3	17.8	14.4	8.1	1.4	-2.0	(31)
(32)			MCDWB	-1.0	-0.6	5.2	12.9	18.7	21.0	23.7	23.1	18.8	11.6	2.7	-0.9	(32)
(33)		10%	WB	-4.6	-3.9	0.8	5.2	11.1	15.2	17.5	16.9	13.2	6.7	0.1	-4.1	(33)
(34)			MCDWB	-3.7	-2.8	3.3	10.2	17.1	19.8	22.5	21.6	17.0	9.5	1.2	-3.4	(34)
(35)	Mean Daily Temperature Range	MDBR	9.0	10.2	11.2	10.7	11.2	9.7	9.5	9.0	8.5	6.8	6.2	7.8	(35)	
(36)		5% DB	MCDBR	10.9	11.7	12.2	14.6	14.8	13.1	12.7	12.3	12.5	11.0	7.5	9.6	(36)
(37)			MCWBR	9.4	9.6	8.2	8.3	7.1	5.7	5.5	5.6	6.6	6.4	5.5	8.6	(37)
(38)		5% WB	MCDWR	10.8	11.2	11.5	14.0	12.9	10.4	11.0	11.1	11.3	10.3	7.0	9.6	(38)
(39)	MCWBR		9.5	9.3	7.9	8.2	6.6	5.2	5.4	5.5	6.5	6.4	5.4	8.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.231	0.261	0.271	0.315	0.331	0.357	0.368	0.348	0.310	0.284	0.238	0.216	(40)	
(41)		taud	2.318	2.309	2.410	2.459	2.476	2.420	2.411	2.459	2.563	2.573	2.427	2.282	(41)	
(42)		Ebn at Noon	727	828	914	912	912	887	867	865	862	806	722	656	(42)	
(43)		Edh at Noon	48	73	85	95	100	107	106	95	74	57	43	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.76	1.76	3.34	4.76	5.58	5.59	5.48	4.53	3.10	1.68	0.72	0.49	(44)	
(45)		RadStd	0.07	0.10	0.20	0.33	0.56	0.47	0.34	0.38	0.35	0.26	0.10	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page